Dear Applicant,

Thank you for applying for the **Senior Developer** position at **3dEYE**.

As part of our selection process, we kindly ask you to complete a **test assignment**. This assignment is designed to assess your approach to **memory management, multithreading/concurrency, and performance optimization**. There is no single correct solution, but we are particularly interested in how you tackle the problem.

Once completed, please submit your solution as a **GitHub repository link** to **hr@3deye.me**.

Please write clean, structured, testable, and easy-to-read code. We value simplicity, correctness, maintainability, and sound engineering decisions over unnecessary complexity or premature optimization.

We'd like to see:

- clean, maintainable code
- clear structure and architecture
- logical separation of responsibilities
- testable design and code that can be easily verified
- practical technical decisions
- code that is easy to read, extend, and reason about
- thoughtful handling of edge cases

Advanced or unusual approaches — such as memory-mapped files, custom sorting strategies, SIMD, or other low-level optimizations — are welcome only as a bonus. They do not compensate for poor fundamentals.

If the solution is not clean, correct, testable, and maintainable, advanced techniques will not be considered an advantage. Complexity without solid engineering quality is a negative signal, not a strength.

The use of AI tools is allowed and welcome. However, the final result is fully the candidate's responsibility. If we see typical AI-generated issues, such as incorrect edge case handling, overengineered abstractions, inconsistent style, dead or irrelevant code, hallucinated APIs, shallow tests, or code the candidate clearly did not review critically — this will reduce the evaluation score. There is no single "correct" structure for this assignment, but our preferred approach is vertical feature-based organization (vertical slice architecture).

AI can help with implementation, but it cannot replace engineering judgment. Using AI is fine. Submitting unreviewed AI output is not.

Test Assignment

Your task is to process a **large text file** where each line follows the format:

<Number>. <String>

Example Input:

415. Apple

30432. Something something something

1. Apple

32. Cherry is the best

2. Banana is yellow

Sorting Criteria:

1. First, sort by the **string** part (alphabetically).
2. If two lines have the same string, sort by the **number** (ascending).

Expected Output:

1. Apple

415. Apple

2. Banana is yellow

32. Cherry is the best

30432. Something something something

Implementation Requirements

You must develop **two C# programs**:

1. **Test File Generator**
    - Creates a text file of the described format.
    - There must be some number of lines with the same String part.
    - Allows specifying the file size.
2. **Sorter**
    - Sorts the file according to the given criteria.
    - Must handle **very large files (~100GB)** efficiently.

Evaluation Criteria

Your solution will be assessed based on:

- **Code quality** and readability.
- **Performance considerations**. Benchmarking and performance testing will be considered a bonus.
- **Memory management**.
- **Test coverage** to ensure reliability.
- **Multithreading** and concurrency.

If you have any questions, feel free to reach out. We look

forward to reviewing your solution!

Best regards,
3dEYE Hiring Team